

HOAS - Hostel Operations Accountability System

> A comprehensive full-stack web application for streamlined hostel/dormitory management with hierarchical role-based access control.

[![Version](https://img.shields.io/badge/version-1.0.0-blue.svg)](https://github.com/nia tappurpose-APPs/HOAS)

[![Firebase](https://img.shields.io/badge/Firebase-v12.6.0-orange.svg)](https://firebase.google.com/)

[![React](https://img.shields.io/badge/React-19.2.0-61dafb.svg)](https://reactjs.org/)

[![Node](https://img.shields.io/badge/Node.js-20+-green.svg)](https://nodejs.org/)

📋 Project Overview

****HOAS (Hostel Operations Accountability System)**** is a modern, enterprise-grade hostel management platform designed for educational institutions. It automates hostel operations from student registration to administrative approvals with a hierarchical permission system spanning four user roles: ****Owner (Super Admin)****, ****Management (Principal/Co-Admin)****, ****Warden****, and ****Student****.

Built with cutting-edge web technologies including ****React 19****, ****Firebase Cloud Functions****, ****Firestore****, and ****Tailwind CSS****, the platform delivers real-time data synchronization, intelligent approval workflows, and role-specific dashboards for efficient hostel administration.

🎯 Project Objectives

- ****Streamline Hostel Management****: Automate and digitize hostel operations end-to-end
- ****Hierarchical Access Control****: Implement granular permission systems with approval workflows
- ****Real-Time Synchronization****: Provide instant updates across all interfaces using Firestore listeners
- ****Scalability****: Support multiple colleges/hostels under unified super admin control
- ****Modern UX****: Deliver intuitive, professional interfaces tailored to each user role

✨ Features

Authentication & Authorization

- 🗝️ ****Google OAuth Integration**** via Firebase Authentication
- 👤 ****Role-Based Access Control (RBAC)**** with custom claims
- 🔑 ****Hierarchical Permissions**** (Owner → Management → Warden → Student)
- ⌚ ****Approval Workflow System**** for new user registrations
- 🏠 ****Persistent Auth State**** with automatic route protection

Dashboard Features

💎 Owner Dashboard (Super Admin)

- Manage all Management (Principal) users across colleges
- Approve/Deny pending management account requests
- ****Cascade Delete Colleges**** - Remove entire college hierarchies (Management → Wardens → Students)
- Real-time statistics and KPI cards
- Profile management with organization details
- Bulk operations and pagination support
- ****Support Ticket System**** - Centralized management of user-reported issues with resolution tracking and status animations.
- ****Global System Settings**** - Configure system-wide settings including:
 - Global toggles (registration, approvals, maintenance mode)
 - Role permission templates with customizable permissions
 - Approval workflow configuration
 - User limits per college/hostel with capacity management
 - Maintenance mode with custom messages

💎 Management Dashboard (Principal/Co-Admin)

- View and manage Wardens and Students linked to specific college
- Approve/Deny pending Warden/Student requests
- KPI cards with pending approvals counter
- Quick approval panel for immediate actions
- Status visualization with circular progress indicators
- Recent activity timeline
- Glassmorphism UI with purple-blue gradient theme

Warden Dashboard

- Monitor assigned students and hostel operations
- AI-powered translation system (multilingual support)
- Update student statuses and manage hostel activities
- Real-time notifications and alerts

Student Dashboard

- View hostel information and personal profile
- Submit requests and view approval status
- Access hostel rules and announcements

Additional Features

- 🎨 ****Modern UI/UX****: Dark theme with glassmorphism effects, smooth animations
- 📊 ****Report Generation****: Export JSON and PDF reports for analytics
- 🔔 ****Toast Notification System****: Beautiful themed notifications with 4 types (Success, Error, Warning, Info)
- 🌐 ****Internationalization****: Multi-language support with i18next
- 🗺️ ****Interactive Dashboard Tour****: Onboarding tutorial using Shepherd.js
- 🌓 ****Theme Toggle****: Auto-detect system theme or manual dark/light mode
- 📱 ****Fully Responsive****: Mobile-first design with breakpoints for all devices
- ⚡ ****Real-time Updates****: Firestore listeners for instant data synchronization
- 🔍 ****Search & Filter****: Advanced filtering across dashboards
- 📄 ****Pagination****: Efficient data loading with page state preservation
- 🚫 ****Server Offline Detection****: Automatic fallback UI when backend is unavailable

🛠️ Tech Stack

Frontend

Technology	Version	Purpose
-----	-----	-----
React	19.2.0	UI framework with modern hooks
Vite	7.2.4	Lightning-fast build tool
Tailwind CSS	4.1.18	Utility-first CSS framework
React Router	7.10.1	Client-side routing
Firebase SDK	12.6.0	Authentication & Firestore client
Lucide React	0.561.0	Modern icon library
Framer Motion	12.27.5	Animation library
Recharts	3.6.0	Data visualization
i18next	25.8.0	Internationalization
Shepherd.js	14.5.1	User onboarding tours

Backend

Technology	Version	Purpose
-----	-----	-----
Node.js	20+	JavaScript runtime
Firebase Functions	v2	Serverless cloud functions
Firebase Admin SDK	Latest	Server-side Firebase operations
Express.js	-	HTTP middleware (within functions)
Firestore	-	NoSQL real-time database
Firebase Auth	-	Authentication service

Development Tools

- ****ESLint**** - Code linting and quality
- ****Concurrently**** - Run multiple npm scripts

- **Firebase Emulator Suite** - Local development environment

📁 Folder Structure

...

HOAS/

```
├── client/                # Frontend Application (React + Vite)
│   ├── public/           # Static assets
│   ├── src/
│       ├── assets/       # Images, icons, animations
│       ├── components/
│           ├── FirebaseModelIndicator.jsx
│           ├── OwnerServices/    # Owner/Admin components
│               ├── AdminLogin.jsx
│               ├── Avatar.jsx
│               ├── DeleteConfirmModal.jsx
│               ├── GlobalDeleteModal.jsx
│               ├── header.jsx
│               ├── OwnerProfile.jsx
│               ├── Sidebar.jsx
│               ├── StatsCard.jsx
│               └── StatusBadge.jsx
│           ├── Routes/
│               ├── index.jsx    # App routing configuration
│               ├── ServerOffline/    # Offline detection component
│               ├── ThemeToggle/    # Dark/Light mode toggle
│               ├── Toast/        # Notification system
│                   ├── Toast.jsx
│                   └── Toast.css
```

```
| | | | └─ ToastContainer.jsx
| | | | └─ README.md
| | | └─ UserServices/    # User role selection
| | └─ context/
| | | └─ AuthContext.jsx  # Firebase auth state
| | | └─ ModalContext.jsx # Modal management
| | | └─ ThemeContext.jsx # Theme management
| | └─ DashBoards/
| | | └─ Management-Dashboard/
| | | | └─ index.jsx
| | | | └─ ManagementDashboard.jsx
| | | | └─ ManagementDashboard.css
| | | | └─ README.md
| | | | └─ QUICK_START.md
| | | └─ Principal-Dashbord/
| | | | └─ PrincipalDashboard.jsx
| | | └─ Student-DashBoard/
| | | | └─ StudentDashboard.jsx
| | | └─ Warden-Dashboard/
| | | | └─ WardenDashboard.jsx
| | └─ firebase/
| | | └─ cloudFunctions.js # Cloud Functions API client
| | | └─ debugUtils.js
| | | └─ firebaseConfig.js
| | └─ hooks/
| | | └─ useServerStatus.js
| | | └─ useSystemSettings.js # System settings enforcement hooks
| | | └─ useTranslation.js
| | └─ Pages/
| | | └─ Dashboard/
| | | └─ HOME/
```

```
| | | └─ LoginPage/
| | | └─ NotFound/
| | | └─ OwnersDashboard/
| | | └─ ProfilePage/
| | | └─ WaitingApproval/
| | └─ App.jsx
| | └─ App.css
| | └─ main.jsx
| | └─ index.css
| └─ index.html
| └─ package.json
| └─ vite.config.js
| └─ eslint.config.js
|
└─ server/                # Backend (Firebase Functions)
  └─ functions/
    └─ src/
      └─ admin.js          # Admin operations
      └─ collegeManagement.js # College CRUD operations
      └─ config.js         # Configuration
      └─ helpers.js        # Utility functions
      └─ reports.js        # Report generation
      └─ systemSettings.js # Global system settings APIs
      └─ triggers.js       # Firestore triggers
      └─ userManagement.js # User CRUD operations
      └─ utility.js        # Utility endpoints
    └─ index.js            # Functions entry point
    └─ package.json
  └─ firebase.json         # Firebase configuration
  └─ serviceAccountKey.json # Firebase credentials (gitignored)
  └─ setAdmin.js           # Admin setup utility
```

```
|
|
|└─ docs/                # Comprehensive Documentation
|
|  └─ ARCHITECTURE_DIAGRAMS.md
|
|  └─ BACKEND_MIGRATION.md
|
|  └─ CHANGELOG.md
|
|  └─ CLOUD_FUNCTIONS_API.md
|
|  └─ CONTEXT_ARCHITECTURE.md
|
|  └─ CONVERSION_SUMMARY.md
|
|  └─ DEPLOYMENT_CHECKLIST.md
|
|  └─ DOCUMENTATION_INDEX.md
|
|  └─ EMULATOR_QUICK_REFERENCE.md
|
|  └─ FIREBASE_EMULATOR_SETUP.md
|
|  └─ FIREBASE_FUNCTIONS_DEPLOYMENT.md
|
|  └─ FIREBASE_MODE_INDICATOR.md
|
|  └─ HOAS_COMPLETE_DOCUMENTATION.md
|
|  └─ Notification.md
|
|  └─ OAUTH_IMPLEMENTATION_SUMMARY.md
|
|  └─ PROJECT_DOCUMENTATION.md
|
|  └─ QUICK_REFERENCE.md
|
|  └─ QUICK_START.md
|
|  └─ REPORTS.md
|
|  └─ ROUTING_AND_CONTEXT.md
|
|
|└─ package.json          # Root workspace configuration
└─ README.md              # This file
...

---

## 🚀 Setup & Installation
```

Prerequisites

Before you begin, ensure you have the following installed:

- **Node.js** 20 or higher ([Download](<https://nodejs.org/>))
- **npm** or **yarn** package manager
- **Firebase CLI** (`npm install -g firebase-tools`)
- **Git** ([Download](<https://git-scm.com/>))
- A **Firebase project** with Firestore and Authentication enabled

Step 1: Clone the Repository

```
``bash
git clone https://github.com/niatappurpose-APPs/HOAS.git
cd HOAS
``
```

Step 2: Firebase Configuration

2.1 Configure Firebase Project

Edit `.firebaseerc` and replace with your Firebase project ID:

```
``json
{
  "projects": {
    "default": "your-firebase-project-id"
  }
}
``
```

2.2 Add Service Account Key

1. Go to Firebase Console → Project Settings → Service Accounts
2. Click "Generate New Private Key"
3. Save the downloaded JSON file as `server/serviceAccountKey.json`

```
```bash
```

```
Ensure the file is in the correct location
```

```
cp /path/to/downloaded/key.json server/serviceAccountKey.json
```

```
```
```

2.3 Update Firebase Config in Client

Edit `client/src/firebase/firebaseConfig.js` with your Firebase web app credentials:

```
```javascript
```

```
const firebaseConfig = {
 apiKey: "your-api-key",
 authDomain: "your-project-id.firebaseio.com",
 projectId: "your-project-id",
 storageBucket: "your-project-id.appspot.com",
 messagingSenderId: "your-sender-id",
 appId: "your-app-id"
};
```

```
```
```

Step 3: Install Dependencies

Option A: Install All Dependencies at Once

```
```bash
```

```
npm run install:all
```

```
```
```

Option B: Install Separately

```
```bash
```

```
Install frontend dependencies
```

```
cd client
```

```
npm install
```

```
cd ..
```

```
Install backend dependencies
```

```
cd server/functions
```

```
npm install
```

```
cd ../..
```

```
```
```

Step 4: Set Up Admin User (First-Time Setup)

```
```bash
```

```
cd server
```

```
node setAdmin.js your-email@example.com
```

```
```
```

This creates the first Owner (Super Admin) account with custom claims.

```
---
```

Usage

Development Mode

Option 1: Run Both Frontend and Backend Concurrently

```
```bash
```

```
npm run dev
```

```
```
```

This starts:

- ****Frontend****: `http://localhost:5173`
- ****Firebase Emulators UI****: `http://localhost:4000`

Option 2: Run Separately

****Terminal 1 - Frontend:****

```
```bash
npm run client
or
cd client && npm run dev
```
```

****Terminal 2 - Backend (Firebase Emulators):****

```
```bash
npm run server
or
cd server && firebase emulators:start
```
```

Firebase Emulator Endpoints

| Service | URL |
|----------------|-------------------------|
| Functions | `http://localhost:5001` |
| Firestore | `http://localhost:8080` |
| Authentication | `http://localhost:9099` |
| Emulator UI | `http://localhost:4000` |

Environment Variables

Create `client/.env` for local emulator testing:

```
```env
```

```
VITE_USE_FIREBASE_EMULATOR=true
```

```
```
```

Building for Production

```
```bash
```

```
Build frontend
```

```
npm run build
```

```
Output will be in client/dist/
```

```
Or build from client directory
```

```
cd client
```

```
npm run build
```

```
```
```

Deployment

Deploy Cloud Functions Only

```
```bash
```

```
npm run deploy
```

```
or
```

```
cd server && firebase deploy --only functions
```

```
```
```

Deploy Everything (Hosting + Functions)

```
```bash
```

```
cd server
```

```
firebase deploy
```

```
```
```

Deploy Specific Function

```
```bash
cd server
firebase deploy --only functions:functionName
```
```

Preview Production Build

```
```bash
cd client
npm run preview
Preview at http://localhost:4173
```
```

Pull Requests & Development History

Development Timeline & Major Features

The HOAS project has evolved through systematic development iterations. Below is a chronological summary of merged changes and feature implementations:

Phase 1: Foundation & Authentication (December 2024)

****PR #1: Initial Authentication System****

- Implemented Google OAuth login via Firebase Authentication
- Created persistent auth state with AuthContext
- Set up role-based routing protection
- Added admin authentication via custom claims
- ****Related Files****: `LoginPage/`, `AuthContext.jsx`, `firebaseConfig.js`

****PR #2: Role Selection & Approval Workflow****

- Built new user role selection flow (Student/Warden/Management)
- Implemented college selection dropdown for hierarchical structure
- Created waiting approval page with real-time status listeners
- Added automatic redirect upon approval
- ****Related Files****: `UserServices/userrole.jsx`, `WaitingApproval/`

**Phase 2: Core Dashboard Development (December 2024 - January 2025)**

****PR #3: Owner Dashboard Implementation****

- Created Owner (Super Admin) dashboard with complete user management
- Implemented approval/denial system for Management users
- Added statistics cards showing total, pending, and approved counts
- Built tab filtering (All/Pending/Approved)
- Real-time updates via Firestore onSnapshot listeners
- ****Commit****: `ded9060` - "Complete HOAS user management system with role-based access control"

****PR #4: Cascade Delete & College Management****

- Implemented cascade delete feature for colleges
- Delete Management account removes all associated Wardens and Students
- Added confirmation modal displaying user counts before deletion
- Enhanced data integrity with batch operations
- ****Related Files****: `OwnersDashboard/`, `DeleteConfirmModal.jsx`, `GlobalDeleteModal.jsx`

****PR #5: Owner Profile Page****

- Created dedicated Owner Profile page (`/owner-profile`)
- Profile editing: display name, phone, organization
- Profile photo display from Google OAuth
- Account status and creation date display
- Logout functionality from profile page
- ****Commit****: `6c24483` - "Add Owner Profile page with edit and logout functionality"

- ****Related Files****: `OwnerProfile.jsx`, `header.jsx`

****PR #6: Management (Principal) Dashboard****

- Built Co-Admin dashboard for college Principals
- View and manage Wardens and Students by college
- Approve/Deny pending Warden/Student requests
- Smart sorting with pending requests prioritized
- Tab switching between Wardens and Students
- ****Related Files****: `Principal-Dashbord/PrincipalDashboard.jsx`

**Phase 3: Backend Migration & Refactoring (January 2025)**

****PR #7: Full-Stack Architecture Restructure****

- Reorganized project into client/server/docs structure
- Separated frontend and backend codebases
- Updated build scripts and workspace configuration
- ****Commits****:
 - `62d55c3` - "Restructure: Organize into full-stack architecture"
 - `d2ef263` - "Complete full-stack restructure and code cleanup"
- ****Related Files****: Root `package.json`, workspace restructure

****PR #8: Firebase Cloud Functions Modularization****

- Refactored backend into modular Cloud Functions structure
- Created dedicated modules: admin.js, userManagement.js, collegeManagement.js, reports.js
- Implemented authorization on every backend endpoint
- Added server-side validation and error handling
- ****Commit****: `247e590` - "Refactor Cloud Functions into modular structure"
- ****Related Files****: `server/functions/src/\*`

****PR #9: Report Generation System****

- Implemented JSON and PDF report exports

- Created BackendReports service for data export
- Added download functionality for user analytics
- ****Commits****:
 - `5772112` - "feat(reports): add BackendReports service for JSON/PDF exports"
 - `54d57fd` - "implement report download functionality and concurrent dev scripts"
- ****Related Files****: `server/functions/src/reports.js`, `Pages/Reports/`

****Phase 4: UX Enhancements & Polish (January 2025)****

****PR #10: Toast Notification System****

- Replaced all browser `alert()` calls with modern toast notifications
- Created custom Toast component with 4 types (Success, Error, Warning, Info)
- Gradient backgrounds matching app theme
- Smooth animations with auto-dismiss and progress bar
- ****Commit****: `9ff1888` - "implement toast notifications, routing improvements, and comprehensive documentation"
- ****Related Files****: `components/Toast/`, Toast integration across dashboards

****PR #11: Pagination & Bulk Operations****

- Added pagination support to Owner Dashboard
- Implemented bulk approve/deny operations
- Enhanced UX with loading states and confirmations
- Page state preservation when navigating
- ****Commits****:
 - `b95bc37` - "Add pagination, bulk operations, and enhanced UX to Owner Dashboard"
 - `fd93f41` - "Add page state preservation when navigating to/from Owner Profile"
- ****Related Files****: `OwnersDashboard/ownersdashbord.jsx`

****PR #12: Theme System & Dashboard Tour****

- Implemented auto-detect system theme mode
- Created manual dark/light theme toggle
- Added interactive dashboard tour using Shepherd.js

- Onboarding guide for new users
- ****Commits****:
 - `c4abcca` - "Implement Dashboard Tour and Auto-System Theme Mode"
 - `726f4bf` - "Add customizable theme pickers and apply background in layout"
- ****Related Files****: `ThemeToggle/`, `context/ThemeContext.jsx`

****PR #13: Server Offline Detection****

- Created ServerOffline component with custom UI
- Automatic detection when Firebase backend is unavailable
- User-friendly error messages and retry logic
- ****Commit****: `299350a` - "Changed UI and styles, added Not Found page and Server Offline page"
- ****Related Files****: `ServerOffline/ServerOffline.jsx`, `hooks/useServerStatus.js`

****PR #14: Internationalization (i18n)****

- Integrated i18next for multi-language support
- AI-powered translation system for Warden Dashboard
- Language detection and switcher
- Translation files for multiple locales
- ****Commits****:
 - `43d71ea` - "Add Language Translations and fixed issues, added content in Owner Dashboard"
 - `874b066` - "Implement and fix AI-powered translation system for Warden Dashboard"
- ****Related Files****: `hooks/useTranslation.js`, locale files

**Phase 5: Management Dashboard Redesign (January 2025)**

****PR #15: Premium Management Dashboard****

- Complete redesign with glassmorphism effects
- Dark theme with purple-blue gradient backgrounds
- KPI cards for Wardens, Students, Pending Approvals, Hostels
- Quick approval panel with user avatars

- Recent activity section with horizontal cards
- Status visualization with circular progress indicators
- Fully responsive with mobile-first design
- ****Commit****: `09235c5` - "managemntdaschboard changed"
- ****Related Files****: `Management-Dashboard/ManagementDashboard.jsx`,
`Management-Dashboard/README.md`

**Phase 6: UI/UX Refinements (January 2025)**

****PR #16: Login Page Interface Update****

- Redesigned login page interface
- Updated routes and navigation paths
- Improved user flow and authentication experience
- ****Commit****: `e65f70b` - "the login page interface was changed by me and some routes path were added"

****PR #17: Small Changes & Bug Fixes**** *_(Multiple Commits)_*

- Various UI improvements and bug fixes
- Code cleanup and optimization
- Loader screen enhancements
- Sidebar and header refinements
- App logo additions

**Phase 7: Recent Updates (January 2026)**

****PR #18: Deployment & Hosting Fixes****

- Resolved Firebase hosting deployment errors
- Updated .gitignore for environment files
- Improved OwnersDashboard responsiveness
- ****Commits****:
 - `5cae3cf` - "fix: resolve Firebase hosting deployment error"
 - `848e52b` - "Update OwnersDashboard responsiveness and fix .gitignore"

****PR #19: Documentation Consolidation**** (`15ecb17`)

- Merged all README files into comprehensive root documentation
- Created unified documentation structure

****PR #20: Empty States Enhancement**** (`591867e`)

- Added Empty State components in OwnersDashboard
- Implemented Empty State for Wardens List page

****PR #21: Global System Settings Module**** (`c85afd5`)

- Created a comprehensive Global System Settings module for Owner-only admin panel
- Added global toggles for registration, approvals, and maintenance mode
- Implemented role permission templates with customizable permissions
- Created approval workflow configuration system
- Added user limits per college/hostel with capacity management
- Implemented complete DB schema with Firestore collections
- Built 15+ Cloud Functions APIs for settings management
- Added enforcement logic with React hooks (FeatureGate, MaintenanceGate, RegistrationGate)
- Created useSystemSettings hook for real-time settings enforcement across the app

****PR #22: UI/UX Improvements**** (`99dbc07`)

- Added perfect themes with consistent styling
- Implemented animated videos/graphics for empty states
- Enhanced "Not Found" page visuals

****PR #23: Minor Fixes**** (`4c001d0`)

- Small UI/UX refinements
- Bug fixes and code cleanup

**Phase 8: Advanced Refinement & Authentication (January - February 2026)**

****PR #24: College Profile & Logo System****

- Implemented college logo upload infrastructure for management users
- Added animated profile preview transition with automated redirection
- Enhanced college identity management within the platform
- **Related Files**: `ProfilePage/`, `ManagementDashboard.jsx`

****PR #25: Google OAuth & Auth Flow Optimization****

- Streamlined Google OAuth authentication process
- Improved persistent session handling and role-based initial redirection
- Fixed authentication race conditions during startup
- **Related Files**: `AuthContext.jsx`, `LoginPage/`

****PR #26: Dynamic Homepage & Team Integration****

- Refactored homepage logic for context-aware call-to-action buttons
- Implemented smart "Sign In" vs "Dashboard" visibility based on auth state
- Integrated dynamic developer team data with local image assets
- **Related Files**: `HOME/`, team data configurations

****PR #27: Support Ticketing & Premium Error System** (Today)**

- **Support Ticket Management**: Built a complete Firestore-backed ticketing system for Owners to track and resolve user issues in real-time.
- **Resolution Workflow**: Implemented interactive status management (Mark Resolved, In Progress, Delete) with localized loading animations and automatic dashboard transitions.
- **Dual Reporting Mechanism**: Redesigned `ErrorModal` to offer both instant Firestore submission and a fallback "Report via Email" option.
- **Premium UI Scaling**: Upgraded the error reporting interface to a massive `max-w-4xl` "Dashboard" style with glassmorphism and modern squircle aesthetics.
- **Bug Fixes**: Resolved critical state management errors in `SupportTickets.jsx` and fixed a `getoleLabel` typo in the `Wardens.jsx` management view.
- **Related Files**: `SupportTickets.jsx`, `ErrorModal.jsx`, `Sidebar.jsx`, `Wardens.jsx`, `Routes/index.jsx`

Summary Statistics

- ****Total Commits****: 110+
- ****Active Development Period****: December 2024 - February 2026
- ****Major Features****: 27+
- ****Documentation Files****: 22+
- ****Lines of Code****: 13,000+ (estimated)

Notable Refactoring Events

1. ****Component Modularization**** (`d302b66`): Refactored OwnersDashboard into reusable components (Avatar, StatsCard, StatusBadge)
2. ****Analytics Removal****: Removed analytics page as per requirements, focusing on core management features
3. ****Cloud Functions Migration****: Moved from client-side Firestore writes to secure Cloud Functions architecture
4. ****Global System Settings**** (`c85afd5`): Added comprehensive admin settings module with enforcement hooks

👥 Contributors

This project is developed and maintained by:

- ****niatappurpose-APPs**** - Lead Developer & Architect

Contribution Guidelines

We welcome contributions! To contribute:

1. ****Fork the repository****
2. ****Create a feature branch****

```
```bash
```

```
git checkout -b feature/amazing-feature
```

...

3. **\*\*Make your changes\*\*** with clear, descriptive commit messages

4. **\*\*Test thoroughly\*\*** using Firebase emulators

5. **\*\*Commit your changes\*\***

```
```bash
git commit -m 'feat: Add amazing feature'
```
```

6. **\*\*Push to your branch\*\***

```
```bash
git push origin feature/amazing-feature
```
```

7. **\*\*Open a Pull Request\*\*** with detailed description of changes

### ### Commit Message Convention

Follow conventional commits format:

- `feat:` - New features
- `fix:` - Bug fixes
- `refactor:` - Code refactoring
- `docs:` - Documentation updates
- `style:` - Code style changes (formatting, etc.)
- `test:` - Adding or updating tests
- `chore:` - Maintenance tasks

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### ## License

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## ## 📞 Support & Contact

### ### Getting Help

- 📖 **\*\*Documentation\*\***: See the [`docs/`](docs/) folder for comprehensive guides
- 🐛 **\*\*Bug Reports\*\***: Open an issue on GitHub
- 💡 **\*\*Feature Requests\*\***: Submit via GitHub Issues
- ✉️ **\*\*Email Support\*\***: Contact the development team

### ### Useful Links

| Resource | Link |

|-----|-----|

| Quick Start Guide | [`docs/QUICK_START.md`](docs/QUICK\_START.md) |

| Complete Documentation |  
[`docs/HOAS_COMPLETE_DOCUMENTATION.md`](docs/HOAS\_COMPLETE\_DOCUMENTATION.md) |

| API Reference |  
[`docs/CLOUD_FUNCTIONS_API.md`](docs/CLOUD\_FUNCTIONS\_API.md) |

| Deployment Guide |  
[`docs/FIREBASE_FUNCTIONS_DEPLOYMENT.md`](docs/FIREBASE\_FUNCTIONS\_DEPLOYMENT.md) |

| Architecture Diagrams |  
[`docs/ARCHITECTURE_DIAGRAMS.md`](docs/ARCHITECTURE\_DIAGRAMS.md) |

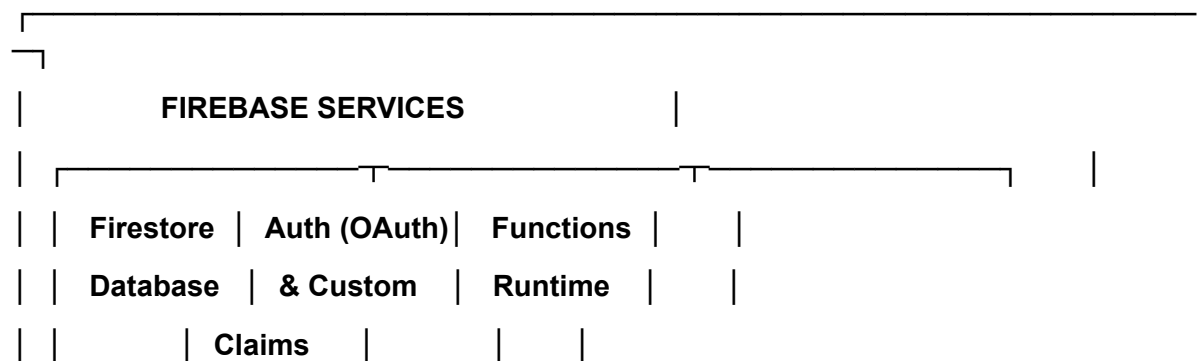
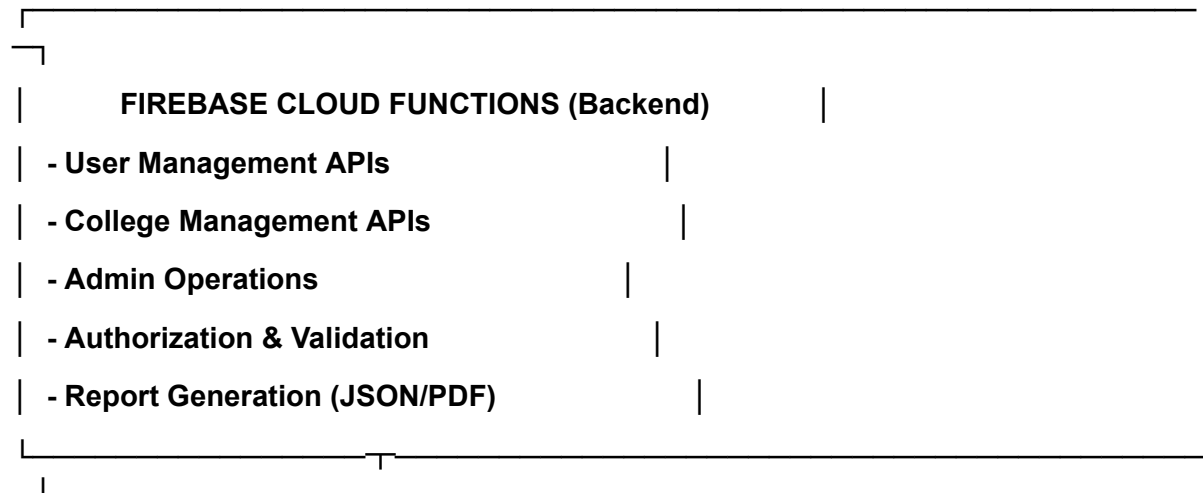
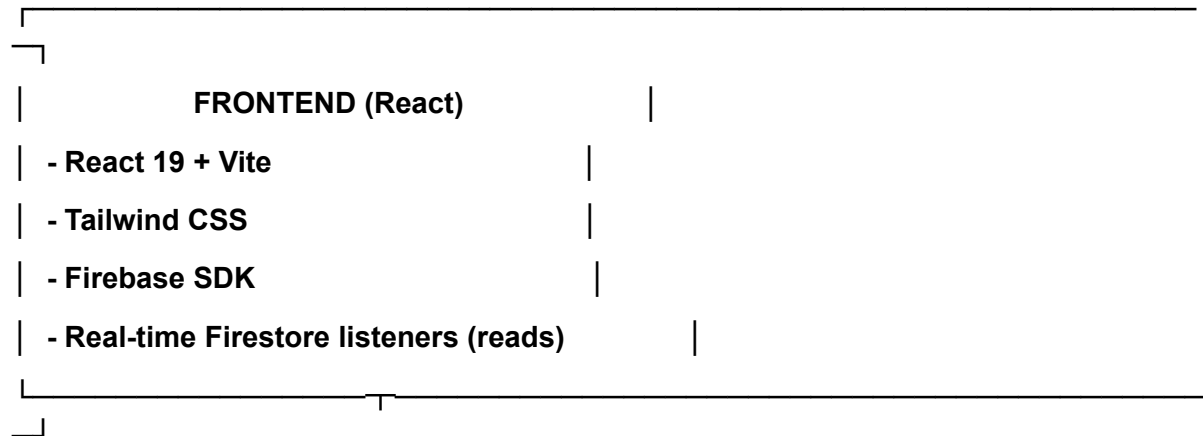
| Changelog | [`docs/CHANGELOG.md`](docs/CHANGELOG.md) |

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## ## 🏗️ Architecture Overview

### ### System Architecture

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### ### Key Color Scheme

| Color  | Hex       | Usage                       |
|--------|-----------|-----------------------------|
|        |           |                             |
| Purple | `#8B5CF6` | Wardens, Primary actions    |
| Blue   | `#3B82F6` | Students, Secondary actions |
| Orange | `#FB923C` | Pending/Attention items     |
| Green  | `#10B981` | Approved/Success states     |
| Red    | `#EF4444` | Deny/Logout actions         |

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### ## 🌟 Future Enhancements

- [ ] Email notifications for approvals/rejections
- [ ] SMS integration for critical alerts
- [ ] Advanced analytics and reporting dashboards
- [ ] Bulk import/export functionality (CSV/Excel)
- [ ] Mobile app (React Native)
- [ ] Hostel inventory management
- [ ] Fee payment integration
- [ ] Complaint/Ticket system
- [ ] Room allocation automation
- [ ] Attendance tracking
- [ ] Parent portal
- [ ] Push notifications
- [ ] Audit logging and activity history
- [ ] Advanced search with filters
- [ ] Data export scheduling

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## ## 🙏 Acknowledgments

- Firebase for providing excellent backend infrastructure
- React community for continuous innovation
- Tailwind CSS for making styling effortless
- Lucide for beautiful, consistent icons
- All open-source contributors whose libraries power this project

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**\*\*Built with ❤️ for Educational Institutions\*\***

**\*\*HOAS - Simplifying Hostel Management, One Dashboard at a Time\*\***

[↑ Back to Top](<#hoas---hostel-operations-accountability-system>)

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